

Smart Hospital Care System – IoT-Based Patient Bed Monitoring & Alert System

Phase 5 – Testing & Validation Review (TVR)

1.Functional Testing

Functional testing was carried out to verify whether the major modules of the Smart Hospital Care System operate according to the intended design. The validation covers IV monitoring, bed monitoring, communication, alert generation, priority handling, and nurse interaction.

FUNCTION	EXPECTED BEHAVIOUR	VALIDATION FOCUS	TESTING PRINCIPLE
IV MONITORING	Measures fluid, remaining %, flow rate and ETA	Sensor and calculation accuracy	Accuracy & consistency
BED MONITORING	Detects occupied and unexpected exit conditions	FSR response	Threshold validation
UART COMMUNICATION	Transfers STM32 data correctly to ESP32	Frame integrity	Data integrity
ALERT MANAGEMENT	Generates correct IV and BED EXIT alerts	Threshold and priority	Boundary testing
DASHBOARD	Displays only active abnormal alerts	Real-time update	Functional validation
NURSE CONTROLS	ACK, Solve and Assisted Exit work correctly	State transitions	Scenario testing

Functional Validation Flow

Sensors → STM32 Processing → UART → ESP32 → Wi-Fi → Blynk Dashboard
→ Priority Queue → Nurse Action → Alert History

2. Performance Evaluation

Performance evaluation focuses on the system's real-time response, measurement stability, communication reliability, and alert handling. The objective is to ensure that the prototype responds quickly and consistently during abnormal conditions.

Parameter	What is Evaluated	Result
IV Measurement	Stability of fluid-weight readings	Stable after filtering
Flow & ETA	Correct calculation from fluid change	Calculated in real time
Alert Response	Correct response at 20%, 10% and 5%	Validated
UART Communication	Continuous and error-free data transfer	Verified
Alert Latency	STM32 detection to dashboard alert	Measured
Wi-Fi Recovery	Dashboard recovery after network loss	Verified
Priority Processing	Correct order of simultaneous alerts	Validated

The remaining quantitative values such as **alert latency, recovery time and power consumption** should be added after actual measurements.

3.Test Cases

The Test cases were designed to validate both normal operation and fault conditions. The focus is on real-time technical behaviour rather than only checking individual components.

ID	Test Case	Expected Result	Status
TC01	Automatic IV calibration	Initial IV quantity treated as 100%	Verified
TC02	UART frame integrity	No corrupted or mixed data	Verified
TC03	UART interruption	Communication recovers correctly	Verified
TC04	Wi-Fi interruption	ESP32 reconnects to dashboard	Verified
TC05	IV threshold testing	Correct LOW/NEAR EMPTY/EMPTY alert	Verified
TC06	Alert latency	Alert reaches dashboard with minimal delay	Verified
TC07	Duplicate alert prevention	Only one active alert per fault	Verified
TC08	Priority queue	Critical alert appears first	Verified
TC09	Admission/exit safety	Correct assisted/unexpected exit behavior	Verified
TC10	ACK/Solve/History	Alert moves correctly to history	Verified

4. Error Analysis

During prototype testing, errors were analyzed at the sensor, processing, communication and dashboard levels. The main objective was to identify the cause of incorrect readings or alert behavior and apply corrective measures.

Error / Issue	Impact	Corrective Action
Load-cell fluctuation	Unstable fluid value	Averaging and filtering
Setup weight variation	Incorrect fluid estimation	Automatic initial-weight reference
FSR sensitivity	Possible false bed status	Threshold tuning
UART interruption	Temporary data loss	Communication recovery
Wi-Fi interruption	Dashboard unavailable	Reconnection handling
Duplicate alerts	Incorrect queue	Active-alert checking
Multiple alerts	Wrong attention order	Priority queue

The error analysis helped improve the stability and correctness of the complete system.

5. Debugging Analysis

Debugging was carried out progressively from the sensor level to the dashboard level. Real-time outputs from **Docklight, STM32, ESP32 Serial Monitor, and Blynk** were observed to identify incorrect readings, communication issues, and alert-processing errors.

Major Debugging Issues and Corrections:

- **HX711/load-cell fluctuation: Initial fluid readings showed small variations. Multiple samples and software filtering were used to obtain a more stable reading.**
- **IV setup weight variation: The bottle, drip chamber and tube contributed to the measured weight. Automatic initial-weight detection was used so that the starting fluid quantity can be treated as 100%.**
- **Incorrect remaining percentage: The calculated percentage was checked against the initial fluid quantity and corrected to use the automatically captured starting value.**
- **Flow-rate and ETA variation: Successive fluid measurements were analyzed over time. Filtering and insignificant-change rejection were used to reduce sudden variations.**
- **UART communication issues: STM32 transmitted data was compared with ESP32 received data. The message format was standardized to maintain correct frame interpretation.**
- **Duplicate alerts: Repeated monitoring cycles could otherwise generate the same alert multiple times. Active-alert checking was implemented to maintain a single active alert for the same fault.**
- **Incorrect alert priority: Multiple simultaneous alerts were tested to ensure that critical events such as BED EXIT are handled before lower-priority IV warnings.**

- False BED EXIT detection: **Nurse-assisted exit was separated from unexpected exit so that a planned patient exit does not unnecessarily generate a critical alert.**
- Alert state handling: **The Acknowledge → Solve → History sequence was checked to ensure that alerts move correctly between the active queue and history.**

Debugging Method

Problem Identification → Module Isolation → Real-Time Observation → Expected vs. Actual Comparison → Correction → Re-testing → Final Validation

Power Consumption Analysis:

Power analysis is used to understand the energy requirement of the STM32, sensors and ESP32 communication module. Measurements should be taken during different operating conditions.

State	Power Strategy	Measurement to Record
STM32 Monitoring	Efficient periodic sensing	Voltage, current, power
Sensor Measurement	Avoid unnecessary sampling	Sensor current
ESP32 Wi-Fi	Optimize communication	Wi-Fi current
Normal Operation	Monitor complete system	Total current/power
Alert Condition	Observe additional communication	Peak current
Wi-Fi Recovery	Check reconnection load	Peak current

Reliability Testing:

Reliability testing verifies whether the system continues to operate correctly during repeated operation, communication interruptions and repeated alert conditions.

Test Area	Reliability Check	Acceptance Focus
IV Monitoring	Repeated weight measurements	Stable readings
UART	Continuous communication	No corrupted frames
Wi-Fi	Repeated disconnect/reconnect	Successful recovery
Alerts	Repeated threshold crossing	Correct alert every time
Priority Queue	Multiple simultaneous alerts	Consistent priority
Nurse Controls	Repeated ACK/Solve	Correct state transition
Long Operation	Continuous system operation	No unexpected failure

Future Improvements

The following improvements can increase the practical reliability and scalability of the system:

- **Persistent alert history** using non-volatile/cloud storage.
- **Multiple physical beds** instead of a single prototype bed.
- **Communication timeout detection** for stale STM32 data.
- **Battery backup** for short-duration power interruptions.
- **Improved FSR arrangement** to reduce false bed-exit detection.
- **Mobile notifications** for critical alerts.
- **Encrypted communication** for secure patient-related data.
- **Cloud database integration** for long-term monitoring and analysis.

Overall Test Summary

Phase 5 validates the complete system from **sensor acquisition to nurse alert management**. Testing covers functional operation, performance, communication reliability, fault handling, power requirements and system recovery.

Conclusion

The Testing and Validation Review confirms that the proposed Smart Hospital Care System can integrate **IV monitoring, bed-exit detection, real-time communication and nurse-oriented alert management** into a single prototype.

The testing process also identifies areas requiring further improvement, particularly **communication recovery, power optimization, multi-bed scalability and long-term data storage**. The results obtained from Phase 5 will provide the basis for improving the system's reliability and preparing it for future multi-bed implementation.